

Real Estate Property Management System (REPMS)

Project Overview

Real Estate Property Management System (REPMS) is a Salesforce-based application developed to manage properties, tenants, lease agreements, vendors, maintenance requests, and real estate operations efficiently.

The application was built using Salesforce standard development practices including Apex, Lightning Web Components (LWC), Custom Objects, Automation, Reports, and Dashboards.

Implementation Details

1. Salesforce Data Model

Created custom objects to manage real estate business processes.

Property Management

Object: Property__c

Implemented fields to store:

- Property Name
- Property Type
- Property Status
- Description
- Address Details
- City
- State
- Postal Code
- Rent Details

Features implemented:

- Property availability tracking
 - Property listing management
 - Property search and filtering
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Tenant Management

Object: Tenant__c

Implemented fields:

- Tenant Name
- Phone
- Email

Features implemented:

- Tenant record management
 - Tenant listing
 - Search functionality
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Vendor Management

Object: Vendor__c

Implemented fields:

- Vendor Name
- Phone
- Email

Features implemented:

- Vendor record management
 - Vendor association with maintenance requests
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Lease Agreement Management

Object: Lease_Agreement__c

Implemented relationships:

- Tenant lookup
- Property lookup

Implemented fields:

- Start Date
- End Date
- Monthly Rent

Features implemented:

- Lease tracking
 - Tenant-property relationship management
 - Expiry monitoring
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Maintenance Request Management

Object: Maintenance_Request__c

Implemented relationships:

- Property lookup
- Vendor lookup

Implemented fields:

- Request Description
- Status

Features implemented:

- Maintenance tracking
 - Vendor assignment
 - Status-based monitoring
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2. Apex Development

Developed Apex classes to handle business logic and server-side processing.

Controllers Developed

PropertyController

Implemented:

- Property retrieval
 - Server-side pagination
 - Search functionality
 - Filtering
 - Sorting
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PropertyCreateController

Implemented:

- Property creation
 - Property validation handling
 - Image attachment processing
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ImageUploadController

Implemented:

- File upload functionality
 - ContentVersion handling
 - Linking uploaded images with property records
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TenantController

Implemented:

- Tenant data retrieval
 - Pagination
 - Search by tenant name
 - Sorting
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LeaseAgreementController

Implemented:

- Lease agreement retrieval
 - Pagination
 - Search
 - Sorting
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MaintenanceRequestController

Implemented:

- Maintenance request retrieval
- Status filtering
- Pagination
- Sorting

VendorAssignmentService

Implemented:

- Automatic vendor assignment logic
 - Maintenance request vendor mapping
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3. Lightning Web Components (LWC)

Developed reusable Lightning Web Components for user interaction.

Property List Component

Implemented features:

- Display property records
 - Server-side pagination
 - Search
 - Filters
 - Sorting
 - Loading spinner
 - Toast messages
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Property Create Component

Implemented features:

- Create property records
 - User input validation
 - Image upload integration
 - Success and error toast notifications
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Image Upload Feature

Integrated with Property Creation workflow.

Implementation:

- User enters property details

- Uploads property images
 - Property record is created
 - Images are attached automatically
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Tenant List Component

Implemented features:

- Tenant listing
 - Search
 - Sorting
 - Pagination
 - Loading spinner
 - Error handling
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Lease Agreement List Component

Implemented features:

- Lease agreement listing
 - Tenant and property relationship display
 - Pagination
 - Search
 - Sorting
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Maintenance Request List Component

Implemented features:

- Maintenance request listing
 - Property and vendor information display
 - Status tracking
 - Pagination
 - Search and sorting
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4. Workspace Implementation

Created a unified REPMS workspace to access multiple modules from a single page.

Integrated components:

- Property List
 - Property Create
 - Tenant List
 - Lease Agreement List
 - Maintenance Request List
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5. Reports and Dashboards

Created analytical reports and dashboard components.

Reports Created

Properties Report

Used for:

- Property availability analysis
- Occupied vs Available tracking

Tenants Report

Used for:

- Tenant management analysis

Lease Agreement Report

Used for:

- Lease monitoring
- Expiry tracking

Maintenance Request Report

Used for:

- Maintenance analysis
- Status tracking

Vendor Report

Used for:

- Vendor workload analysis
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Dashboard Components

Implemented dashboard visualizations:

Lease Agreements Expiring in Next 30 Days

Displays upcoming lease expirations.

Maintenance Requests by Status

Displays maintenance distribution:

- Open
 - In Progress
 - Completed
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Occupancy Rate

Displays:

- Occupied properties
 - Available properties
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Available vs Occupied Properties

Provides property availability comparison.

Vendor Workload

Shows maintenance request distribution among vendors.

6. Testing

Implemented Apex test coverage for business logic.

Created:

- Test Data Factory
- Vendor Assignment Service Test

Validated:

- Record creation
 - Business logic execution
 - Automation behavior
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7. Additional Planned Enhancements

The following enhancements were planned for future implementation:

Agentforce Integration

Planned usage:

- AI-based property assistance
- Automated tenant support
- Intelligent maintenance query handling

Experience Cloud

Planned implementation:

- Tenant self-service portal
- Property browsing
- Lease information access
- Maintenance request submission

These enhancements were considered future extensions beyond the current assignment scope.

Technology Stack

- Salesforce Platform
- Apex

- SOQL
 - Lightning Web Components
 - Salesforce Custom Objects
 - Salesforce Reports & Dashboards
 - Salesforce Automation
 - GitHub Version Control
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Conclusion

REPMS demonstrates a complete Salesforce application development lifecycle including data modeling, Apex development, Lightning Web Components, user interface implementation, reporting, dashboards, and testing.